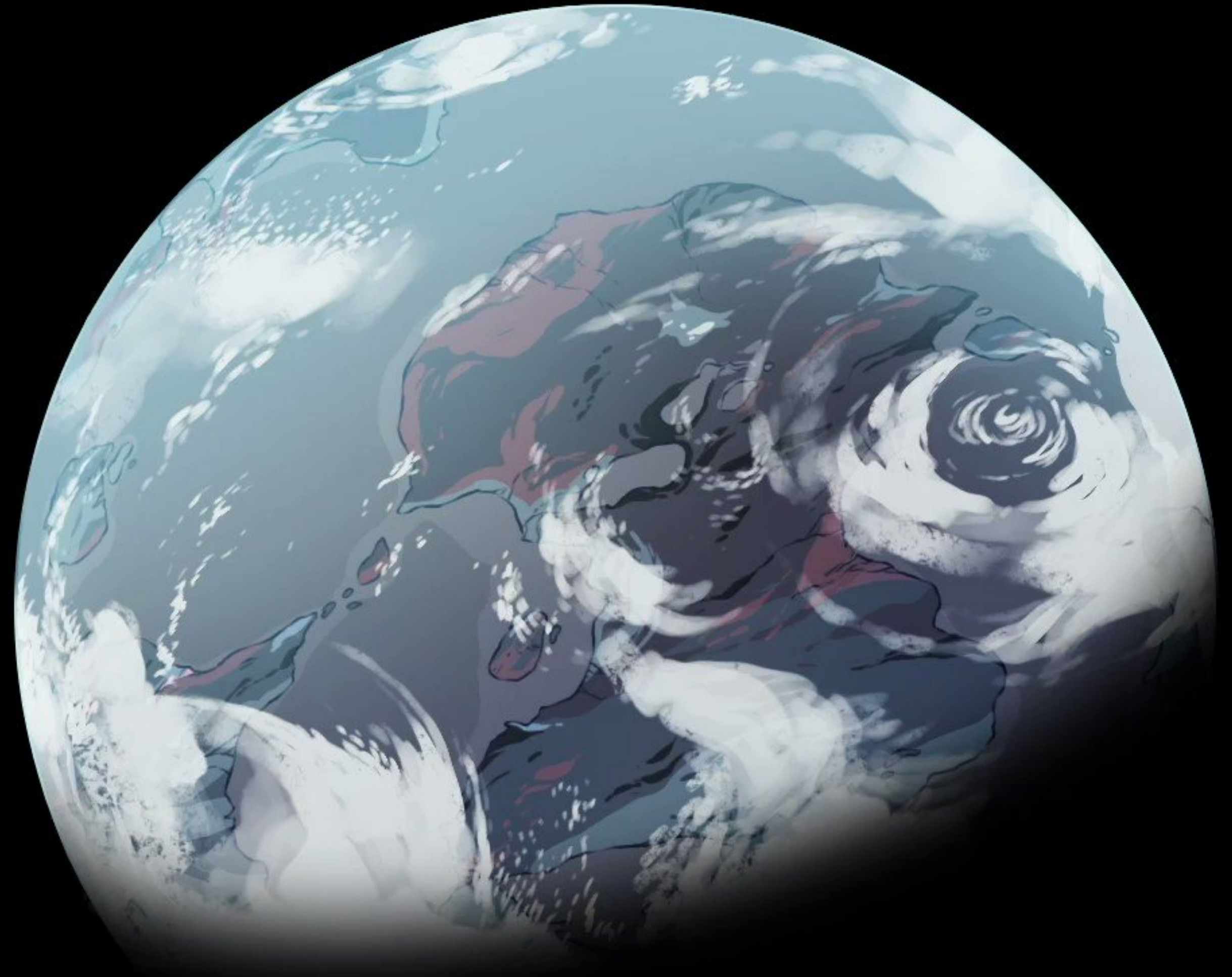


Science & Science Fiction

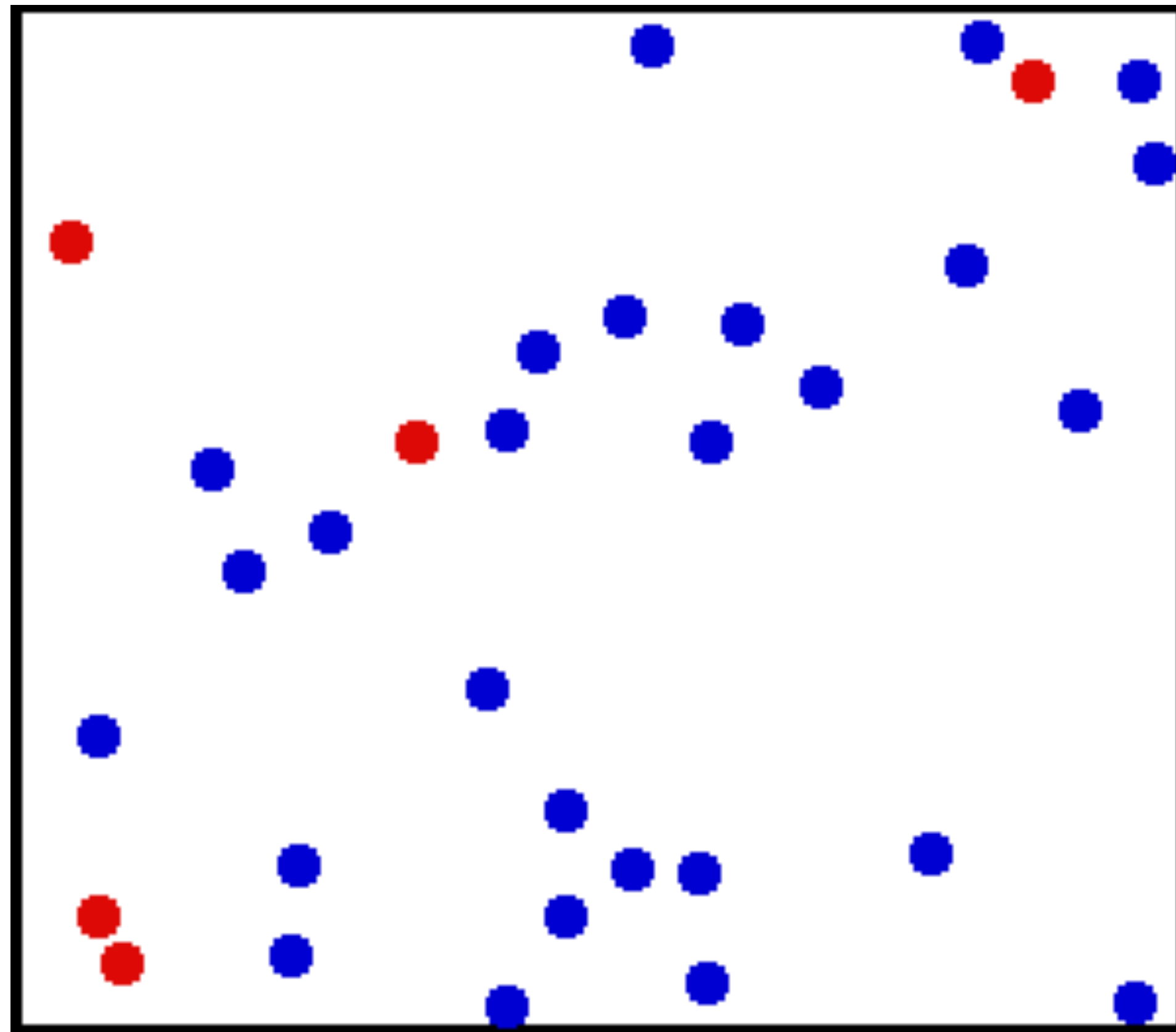
33-120 | Prof. Katie Breivik



Announcements

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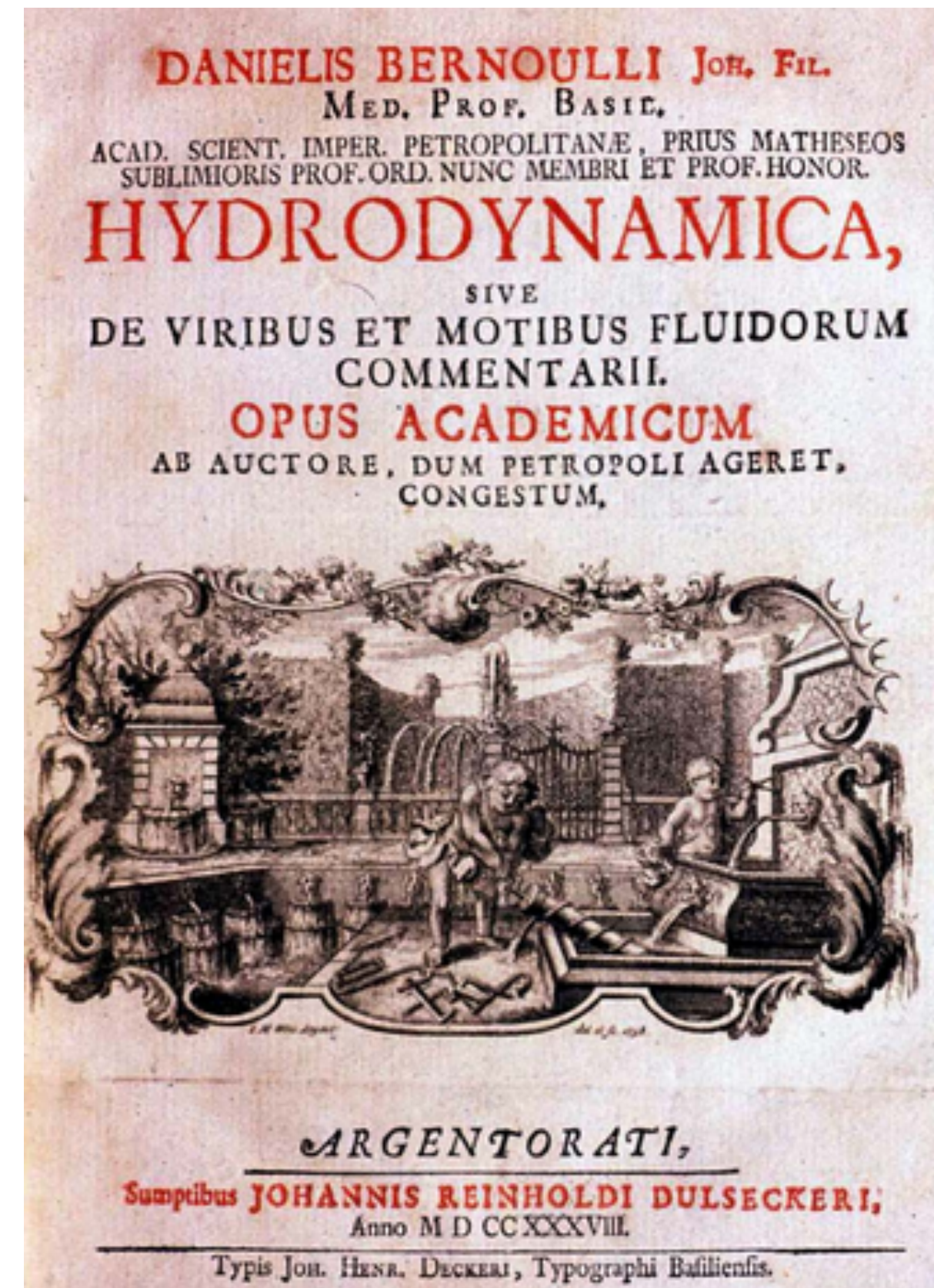
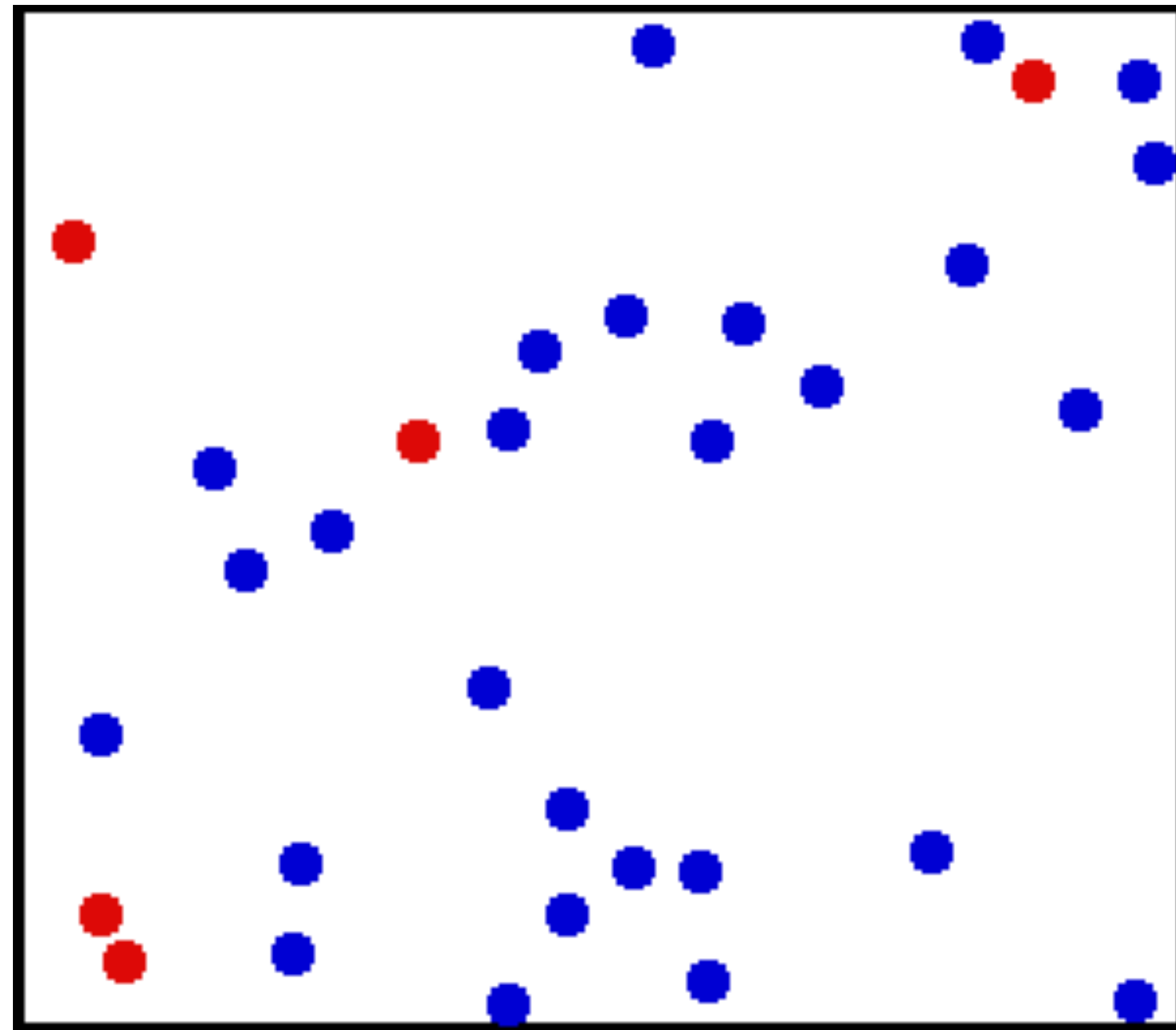
Kinetic theory of gases



ideal gas: a theoretical gas made up of particles that are assumed to not experience *inter-particle reactions*

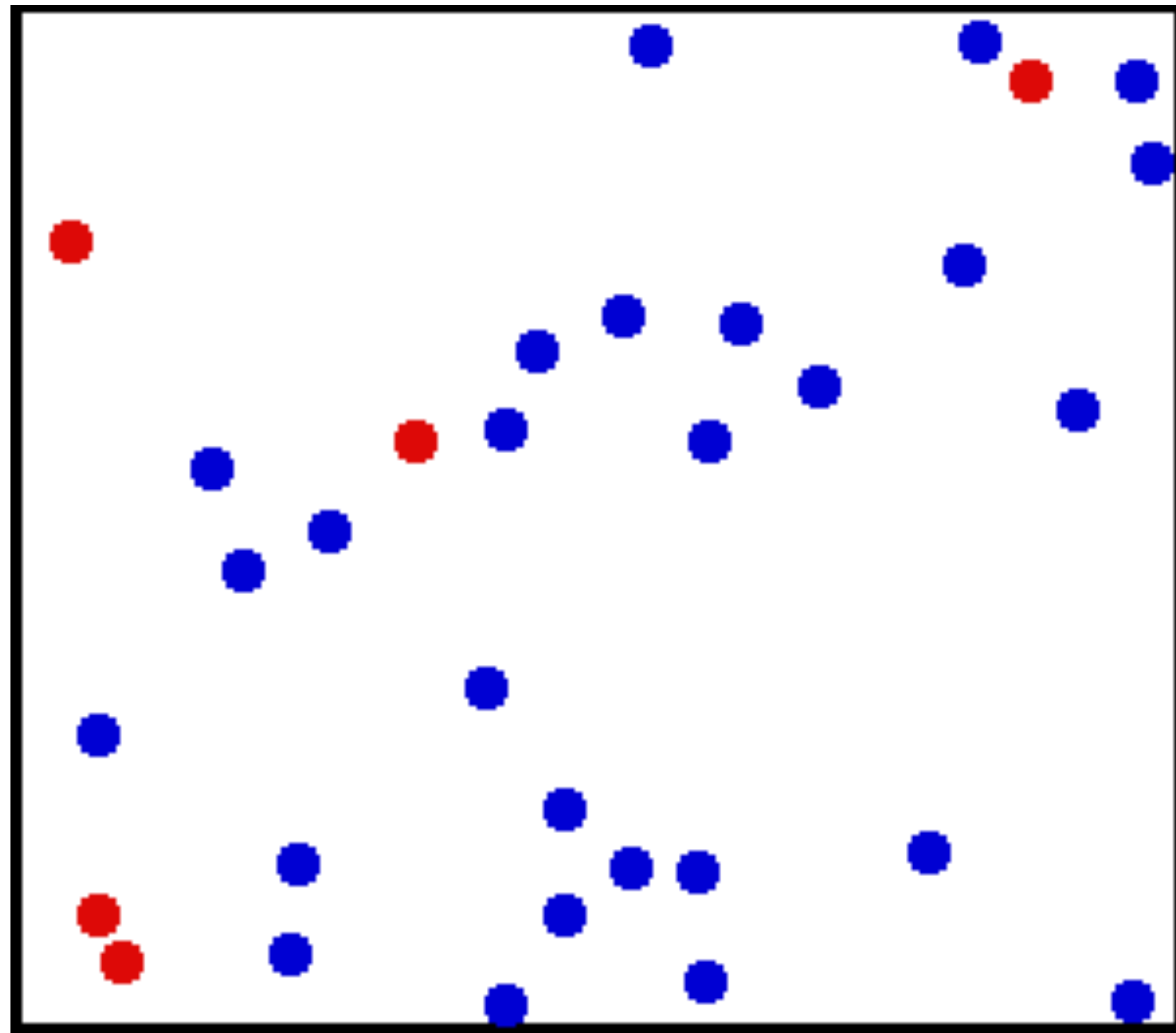
collisions between particles in an ideal gas are used to describe the relationships between the temperature (T), pressure (P), and volume of the enclosure (V) that the gas occupies

Kinetic theory of gases



Pioneered by
Daniel
Bernoulli in
1738 (before
energy
conservation
was
understood!)

Kinetic theory of gases



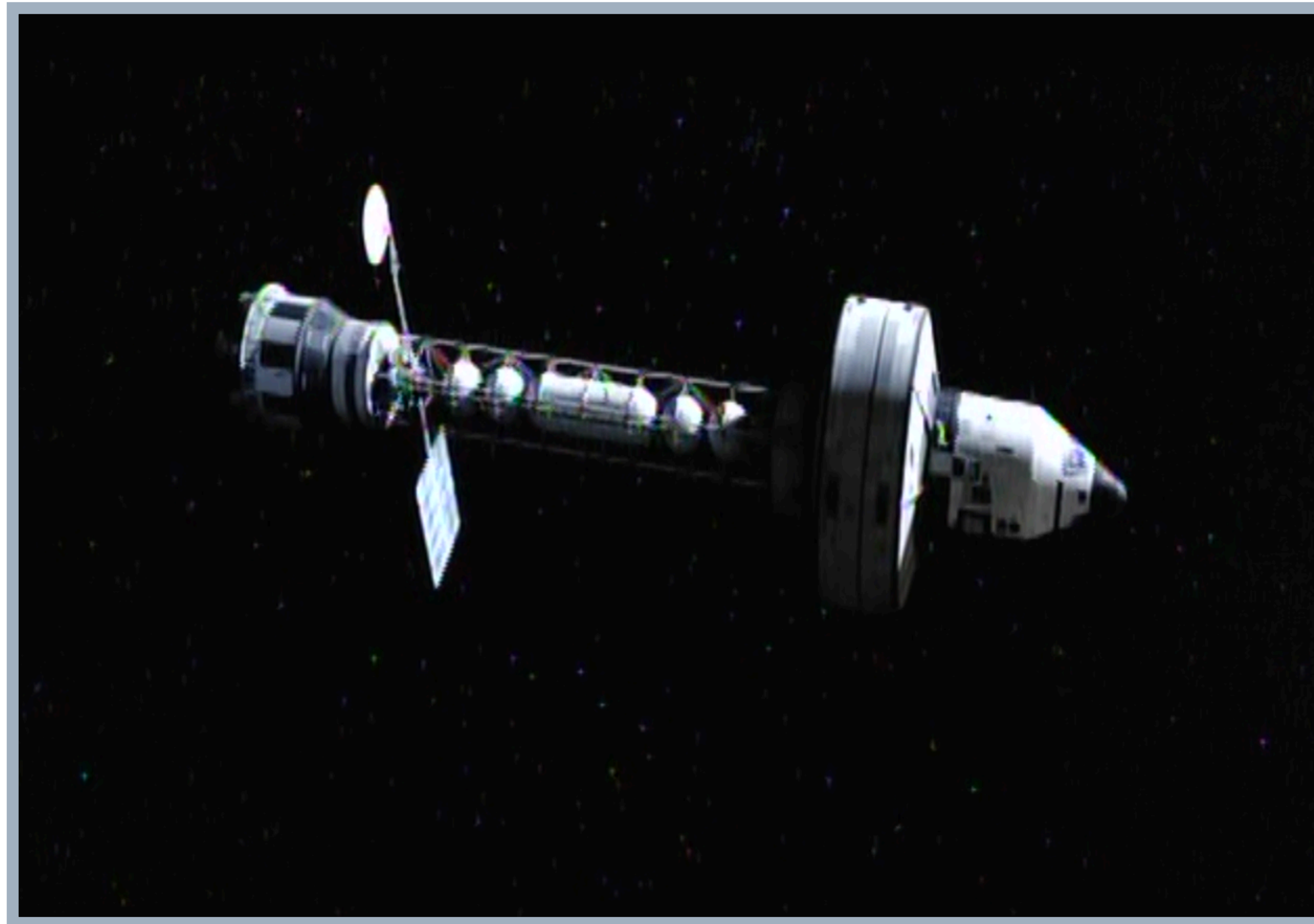
The kinetic energy per particle depends on the temperature of the gas

$$K = \frac{1}{2}mv^2 = \frac{3}{2}k_{\text{B}}T$$

m : mass per particle

k_{B} : Boltzmann's constant

T : temperature of the gas



Mission to Mars

Directed:
Brian De Palma

Touchstone Pictures
(2000)

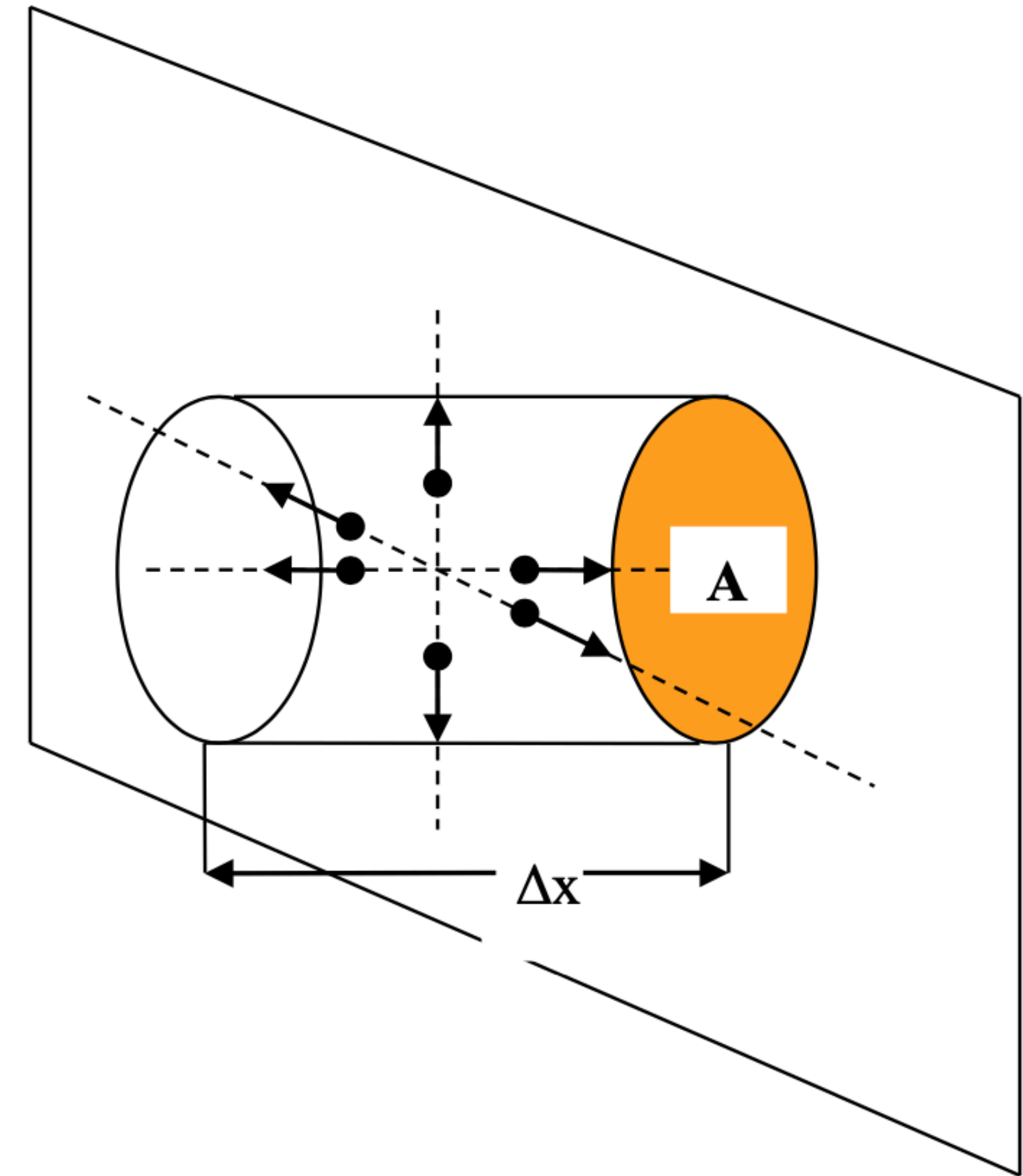
How long would it take to lose 1/2 of the air through a hole?

An approximation:

only 1 of the 6 molecules will be traveling in the right direction (on average) under the assumption that the spacecraft is a cube with 6 surfaces

of particles moving towards the hole:

$$\Delta n = \frac{n}{6} \frac{A}{V} \Delta x$$



How long would it take to lose 1/2 of the air through a hole?

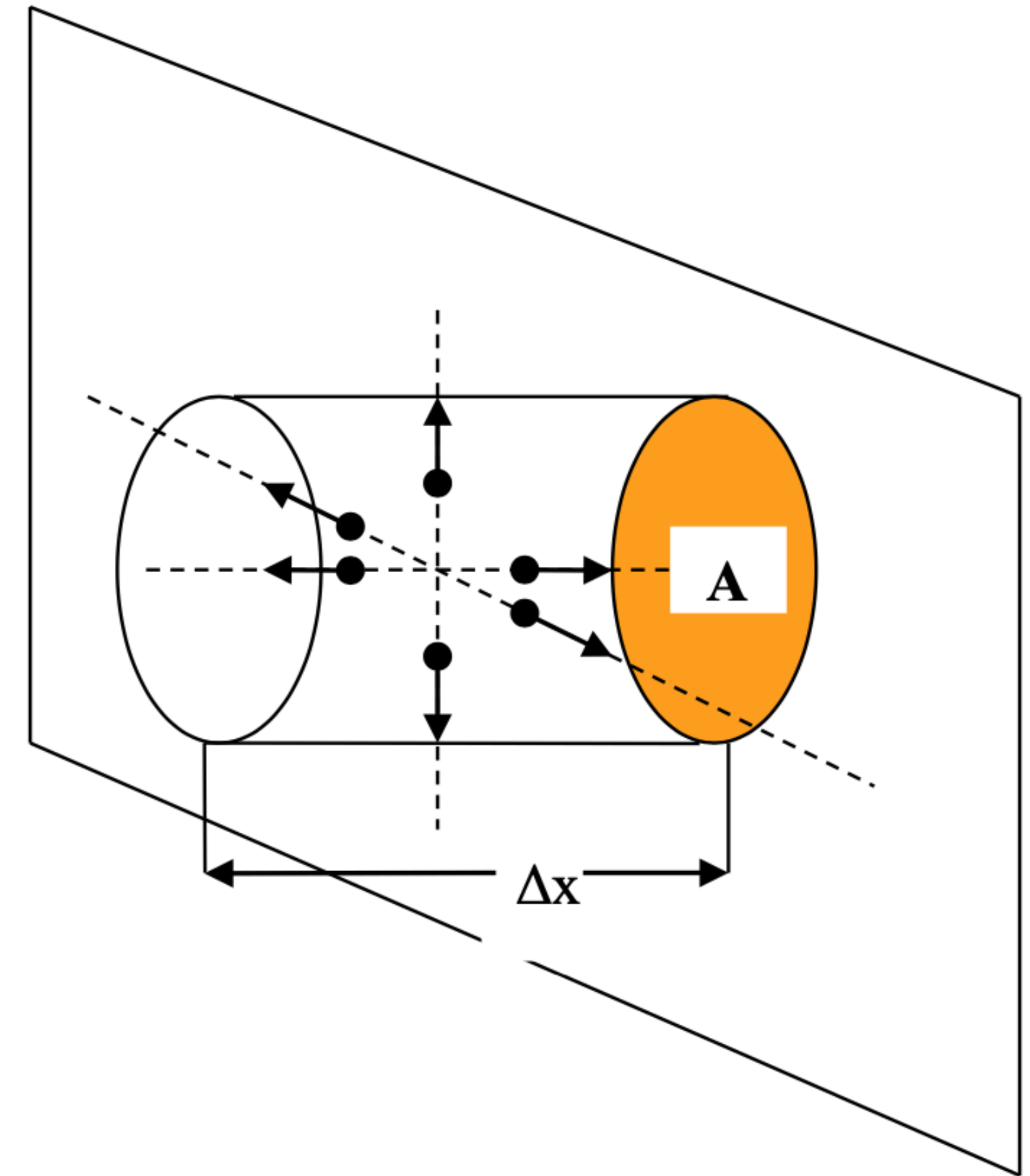
of particles moving towards the hole:

$$\Delta n = \frac{n}{6} \frac{A}{V} \Delta x$$

Speed of each gas particle:

$$v = \Delta x / \Delta t$$

Plug & chug:
$$\Delta n = \frac{n}{6} \frac{A}{V} \Delta x = \frac{n}{6} \frac{A}{V} v \Delta t$$



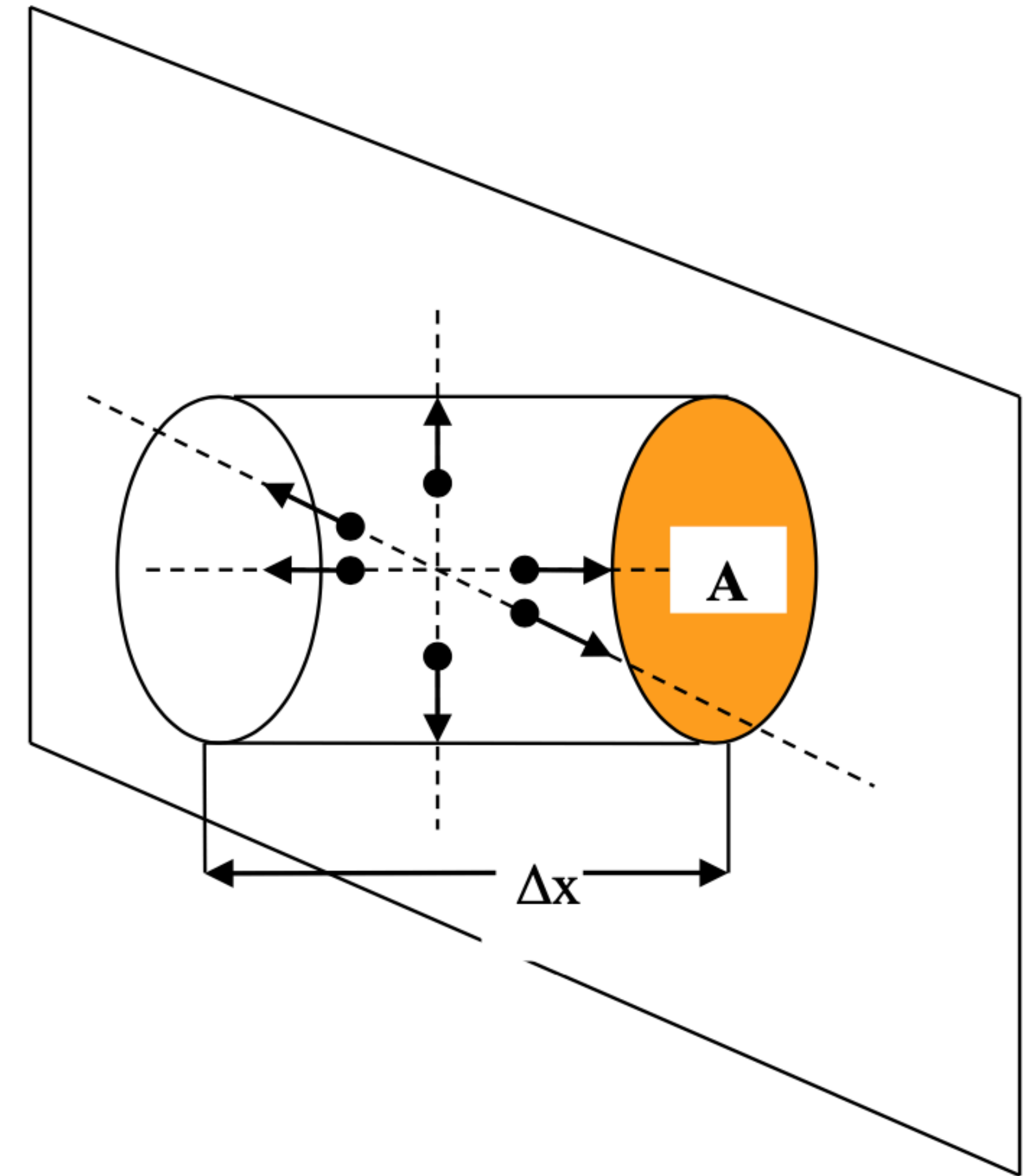
How long would it take to lose 1/2 of the air through a hole?

$$\Delta n = \frac{n}{6} \frac{A}{V} \Delta x = \frac{n}{6} \frac{A}{V} v \Delta t$$

$$\Delta t = 6 \frac{\Delta n}{n} \frac{V}{A v}$$

Recall: $K = \frac{1}{2} m v^2 = \frac{3}{2} k_B T$

$$v^2 = \frac{3}{m} k_B T$$

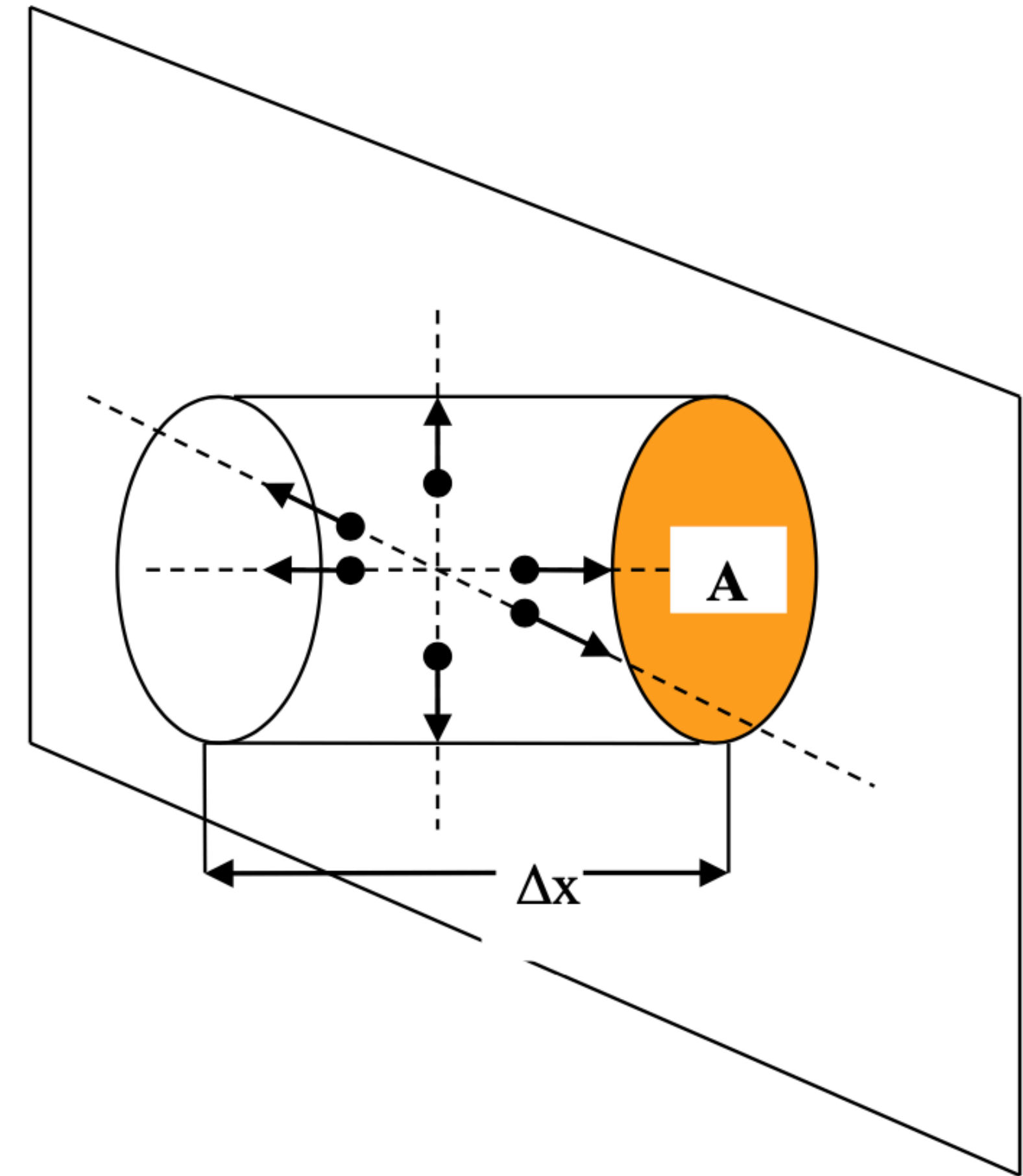


How long would it take to lose 1/2 of the air through a hole?

$$\Delta t = 6 \frac{\Delta n}{n} \frac{V}{A} v \qquad v^2 = \frac{3}{m} k_B T$$

$$\Delta t = 6 \frac{\Delta n}{n} \frac{V}{A} \sqrt{\frac{m}{3k_B T}}$$

1/2 of the air: $\frac{\Delta n}{n} = \frac{1}{2}$



How long would it take to lose 1/2 of the air through a hole?

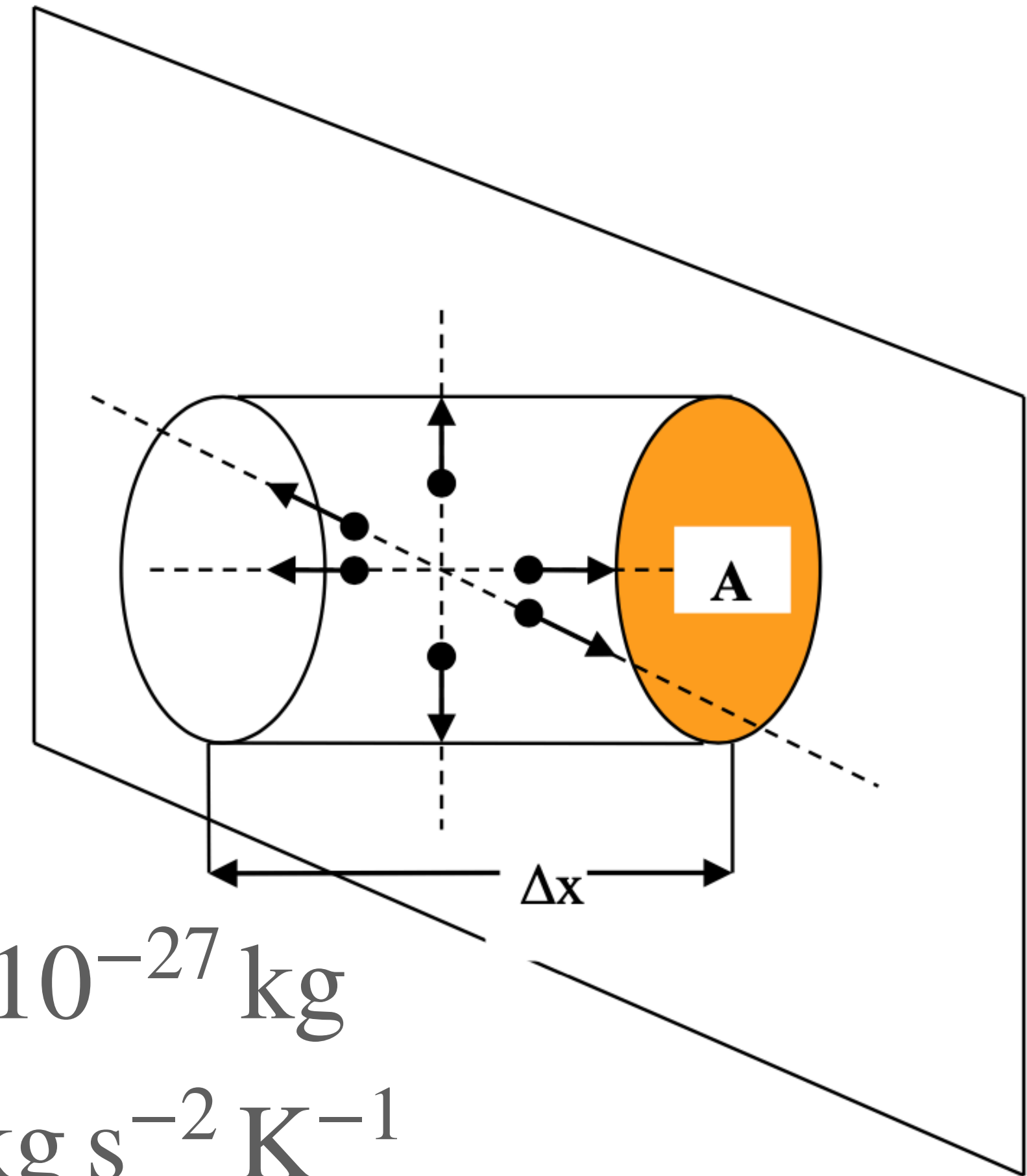
$$\Delta t = 6 \frac{\Delta n}{n} \frac{V}{A} \sqrt{\frac{m}{3k_B T}}$$

$$\Delta t = 3 \frac{V}{A} \sqrt{\frac{m}{3k_B T}}$$

At room temp: $T \simeq 300$ K

For air: $m_{\text{N}_2} = 2 \times 14 \times u = 28 \times 1.66054 \times 10^{-27}$ kg

Boltzmann's constant: $k_B = 1.38 \times 10^{-23} \text{m}^2 \text{kg s}^{-2} \text{K}^{-1}$



How long would it take to lose 1/2 of the air through a hole?

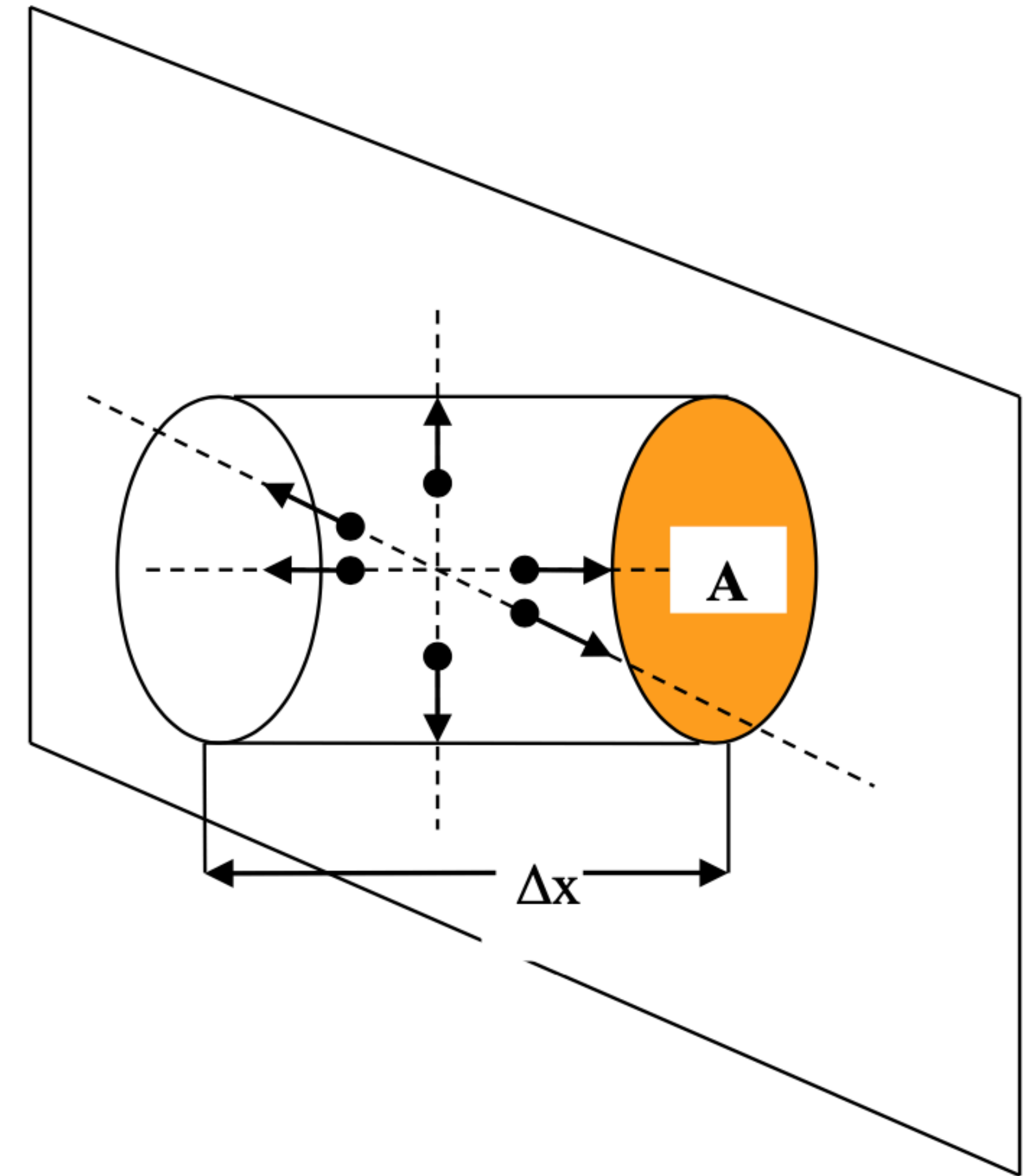
$$\Delta t = 3 \frac{V}{A} \sqrt{\frac{m}{3k_B T}}$$

$$\Delta t = 3 \frac{V}{A} \times 0.002 \text{ s m}^{-1}$$

If the hole is 1cm in diameter:

$$A = \pi \times (0.005 \text{ m}^2) = 0.00008 \text{ m}^2$$

$$\Delta t = 75 \text{ m}^{-3} \text{ s} \times V$$



How long would it take to lose 1/2 of the air through a hole?

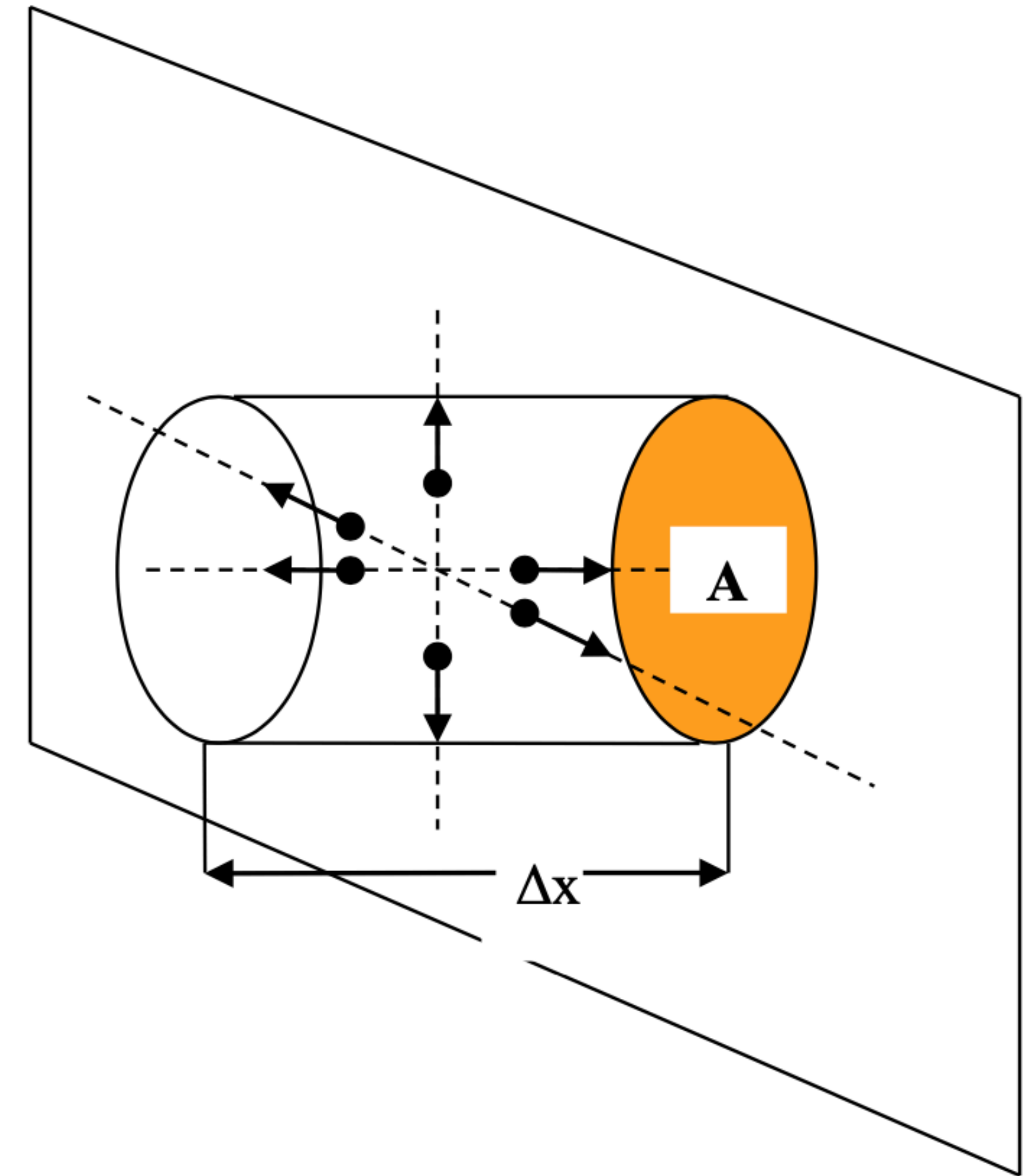
$$\Delta t = 75 \text{ m}^{-3}\text{s} \times V$$

How small does the ship have to be to make the time be ~6 mins?

$$V = \frac{6 \times 60 \text{ s}}{75 \text{ m}^{-3}\text{s}} = 4.8 \text{ m}^3$$

$$R^3 = 4.8 \text{ m}^3 \quad \longrightarrow \quad R = 1.7 \text{ m}$$

What is this? A spaceship for ants?





This did actually happen on August 30, 2018
on the International Space Station!

Problem 4 asks you to repeat the last exercise for the *real thing*

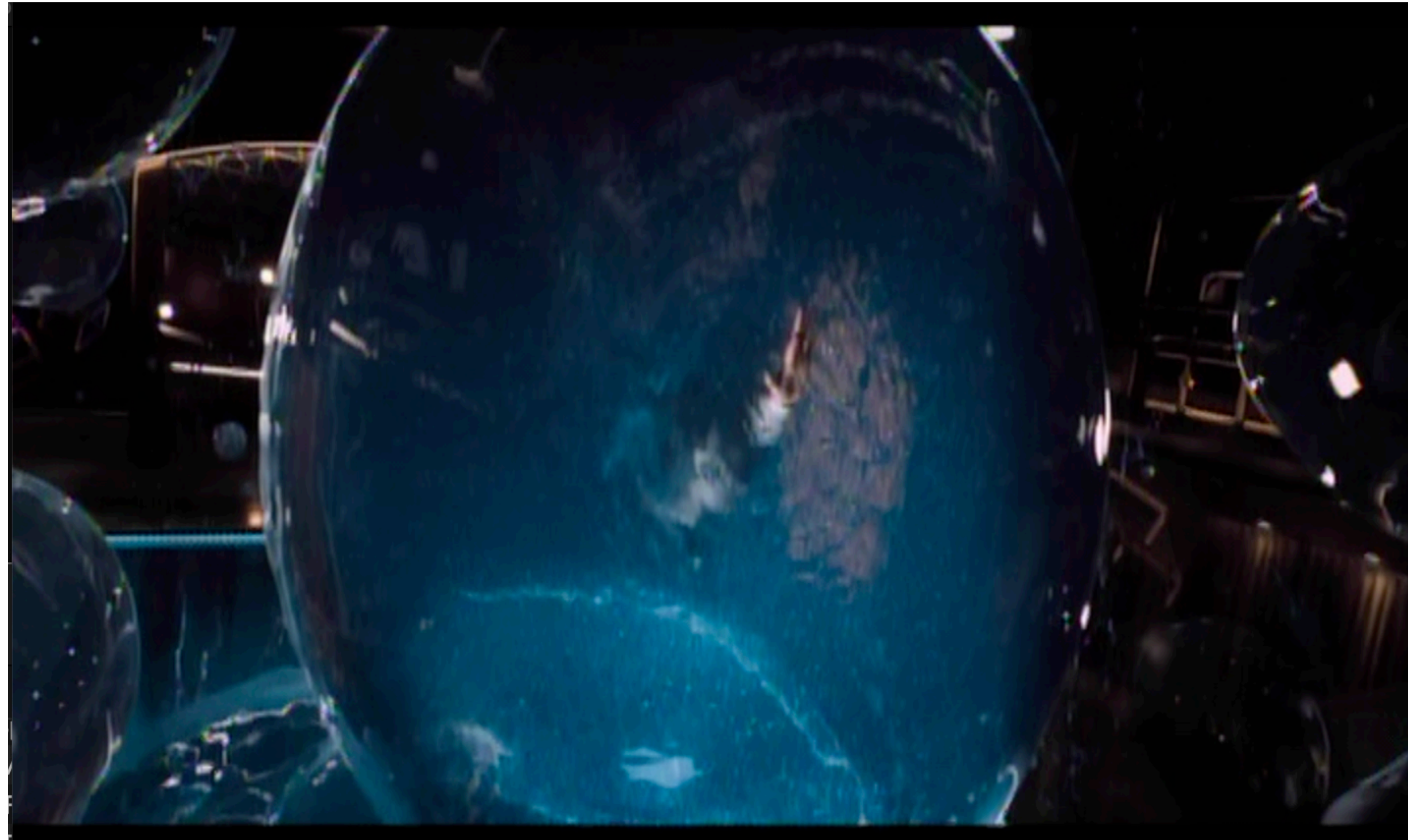


Aliens

Directed:
James Cameron

20th Century Fox
(1986)

Properties of liquids

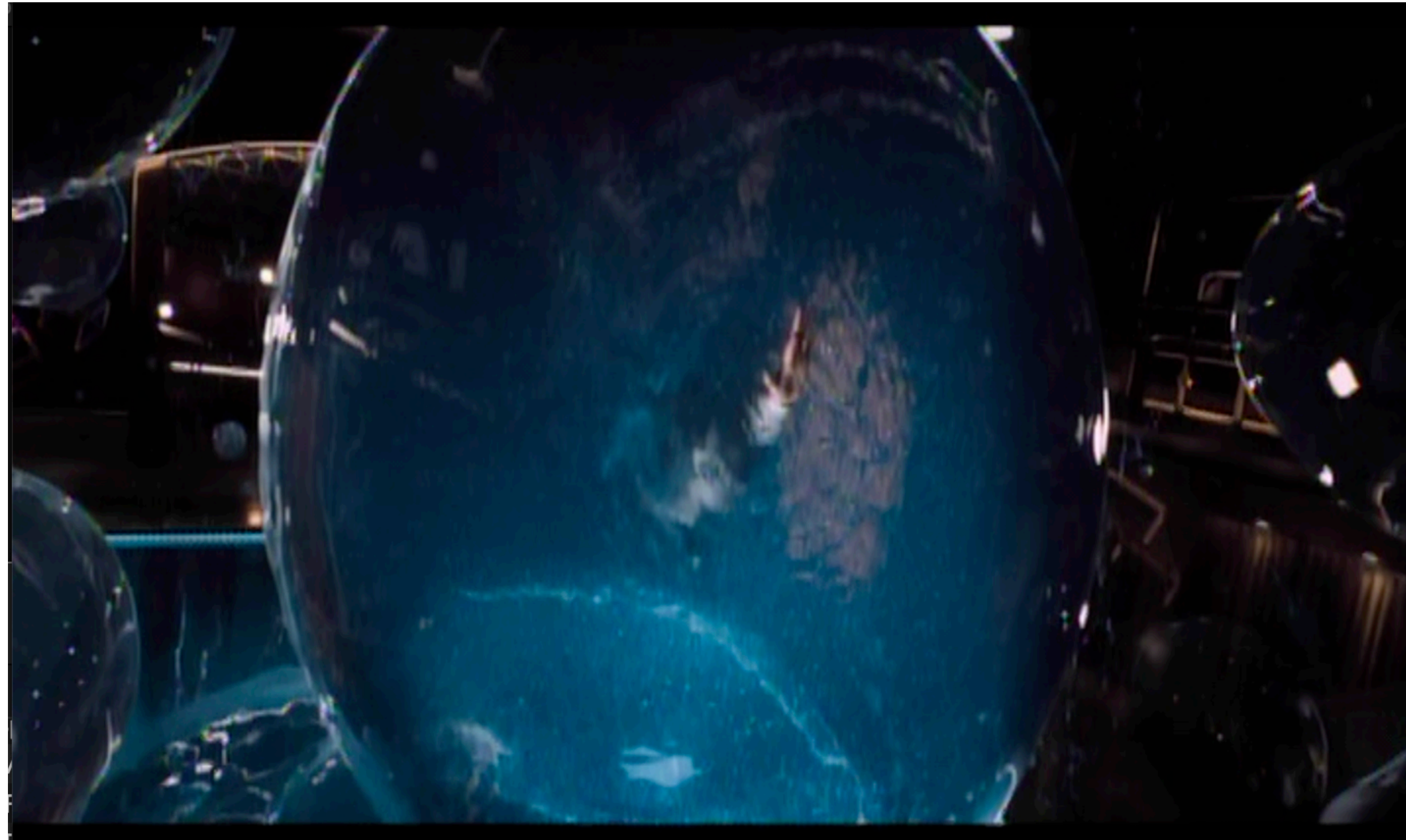


Passengers

Directed:
Moreton Tyledum

Columbia Pictures
(2016)

Properties of liquids



Liquids are subject to:

- ▶ Viscous drag
- ▶ Gravity
- ▶ Surface tension

A condensed matter with no long-range order that preserves the arrangement of atoms or molecules —> always deforms under forces



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